

AJ Weeks

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I am a senior graphics programmer with a strong passion for understanding how things work at their core. I have many years of experience building modern realtime graphics engines with a high quality bar for code architecture and peer review. My strong problem solving abilities make me a fast learner, and I deeply enjoy working in interdisciplinary teams to achieve the best possible result.

Work Experience

Electronic Arts / Rendering Engineer

JAN 2024 – PRESENT, STOCKHOLM, SWEDEN

Primarily responsible for maintaining and improving the surfel-based raytraced global illumination system (GIBS), expanding and tuning the system for each game team's needs. Shipped to millions of players across several major titles.

Technologies used: **Proprietary engine (Frostbite) (C++, HLSL), Perforce, PIX**

Avalanche Studios Group / Experienced Graphics Programmer

JAN 2022 – DEC 2023, STOCKHOLM, SWEDEN

Responsibilities included improving the raytraced global illumination and shadow systems, maintenance of the engine, optimizing CPU & GPU workloads.

Technologies used: **Proprietary engine (Apex) (C++, HLSL), Perforce, PIX**

Electric Square / Programmer

FEBRUARY 2018 - DECEMBER 2021, BRIGHTON, UK

Internship followed by full time employment as a generalist programmer.

Technologies used: **Unity (C#), Unreal Engine 4 (C++, HLSL, Blueprints), Perforce**

Projects

Machine Learning Workshop

2025

Created and led a workshop for 40+ colleagues covering theory and application of ML basics (MNIST digit recognition & real-time image denoiser/upscaler).

Flex Engine

2017 - 2023

Custom game engine (C++, Vulkan). Physically-based renderer supporting features such as tessellated water, procedural terrain, GPU-based particle simulation, screen-space ambient occlusion, cascaded shadow mapping, temporal anti-aliasing, physics solver, audio engine, UI renderer, and much more. ([GitHub](#))

Realtime Global Illumination Research

2019 - 2021

Co-author of unpublished research paper describing novel, scalable, & dynamic global illumination technique. Also responsible for implementation in production.

Education

Digital Arts and Entertainment / Hogeschool West-Vlaanderen

AUGUST 2015 - JUNE 2018, KORTRIJK, BELGIUM

Programming-focused game development BSc. Courses included: Programming, Applied Math & Physics, Unity & UE4, Photoshop. Graduated Magna Cum Laude.